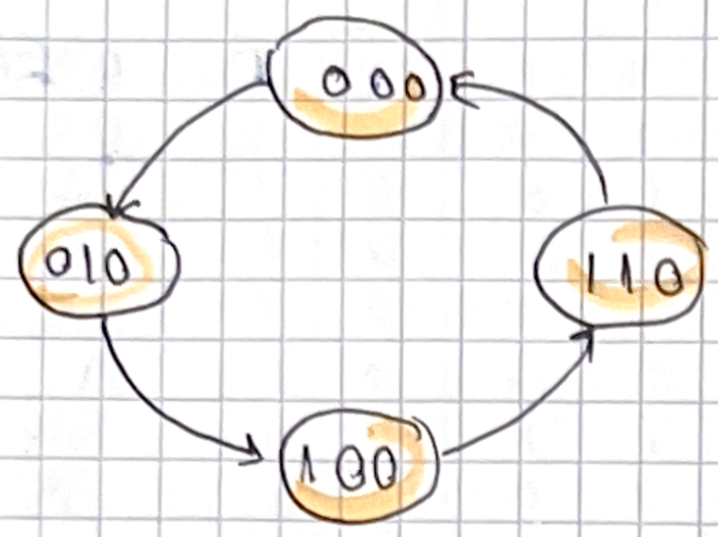
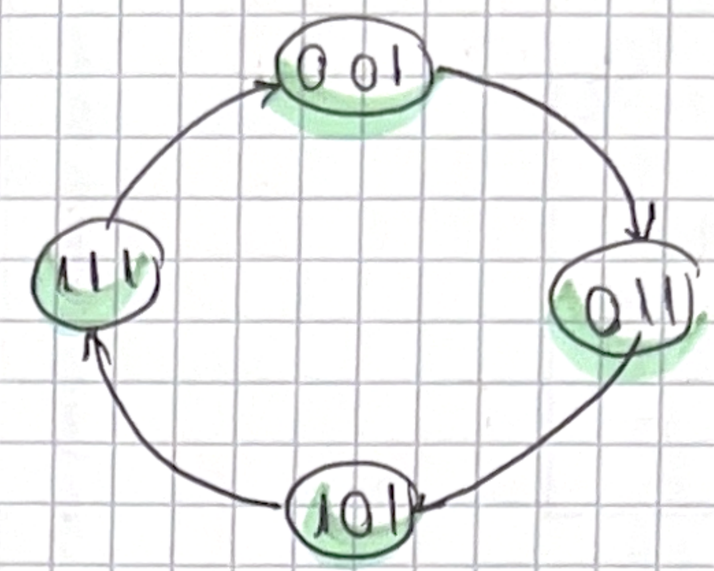
17. 12. 2025.

N_3 BB 0-2-4-6-0 ; 1-3-5-7-1





Q_2	Q_1	Q_0	Q_2^1	Q_1^1	Q^0
0	0	0	0	1	0
0	0	1	0	1	1
0	1	0	1	0	0
0	1	1	1	0	1
1	0	0	1	1	0
1	0	1	1	1	1
1	1	0	0	0	0
1	1	1	0	0	1

$$D_2 = Q_2^1 = \overline{Q}_2 Q_1 + Q_2 \overline{Q}_1$$

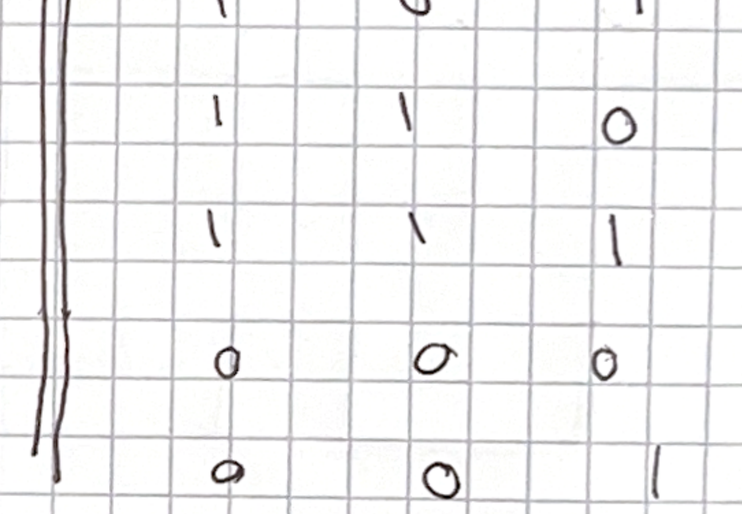
$D_2 = Q_2^1 =$

$= \overline{Q}_0 (\overline{Q}_2 Q_1 + Q_2 \overline{Q}_1) +$

$+ Q_0 (\overline{Q}_2 Q_1 + Q_2 \overline{Q}_1)$

$= \underbrace{(Q_0 + \overline{Q}_0)}_{1} (\overline{Q}_2 Q_1 + Q_2 \overline{Q}_1)$

$= Q_2 \oplus Q_1$



$Q_2 Q_1 \cdot Q_0 +$

$Q_0 \cdot (\overline{Q}_2 Q_1 + Q_2 \overline{Q}_1)$

$+ \overline{Q}_0 \cdot (Q_1 \overline{Q}_2 + Q_1 \overline{Q}_2)$

$$D_1 = Q_1'$$

$$= \bar{Q}_0(\bar{Q}_2\bar{Q}_1 + Q_2\bar{Q}_1) + Q_0(\bar{Q}_2\bar{Q}_1 + Q_2\bar{Q}_1)$$

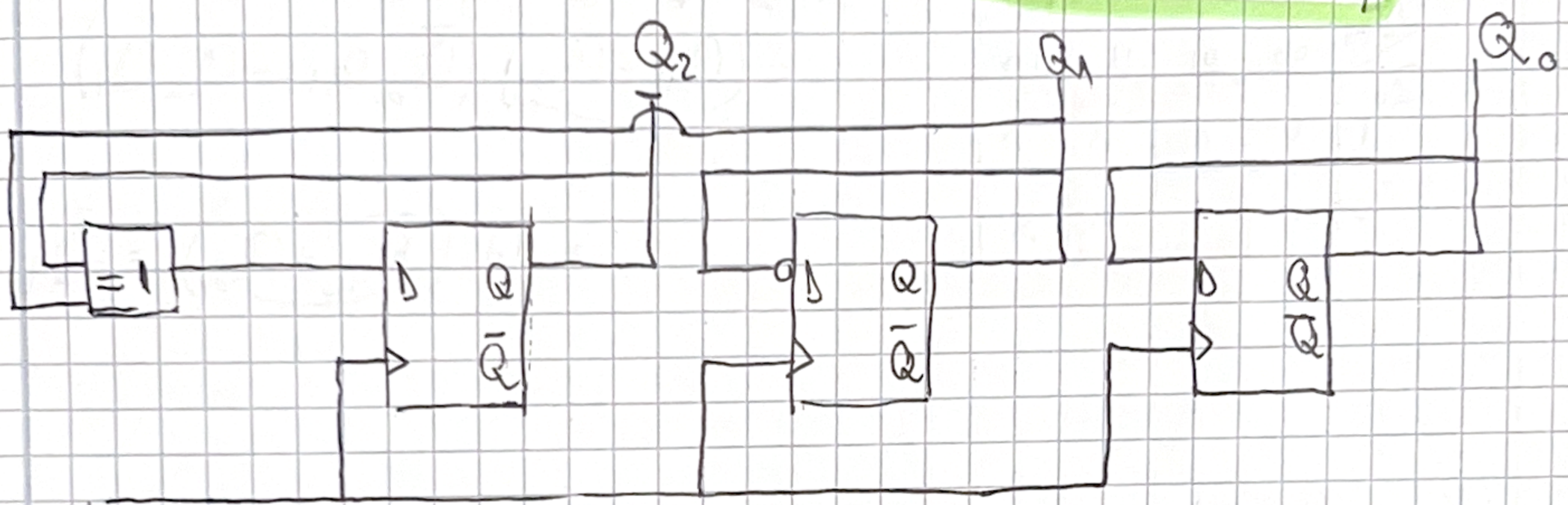
$$= \underbrace{(\bar{Q}_0 + Q_0)}_{1}(\bar{Q}_2\bar{Q}_1 + Q_2\bar{Q}_1)$$

$$= \bar{Q}_1$$

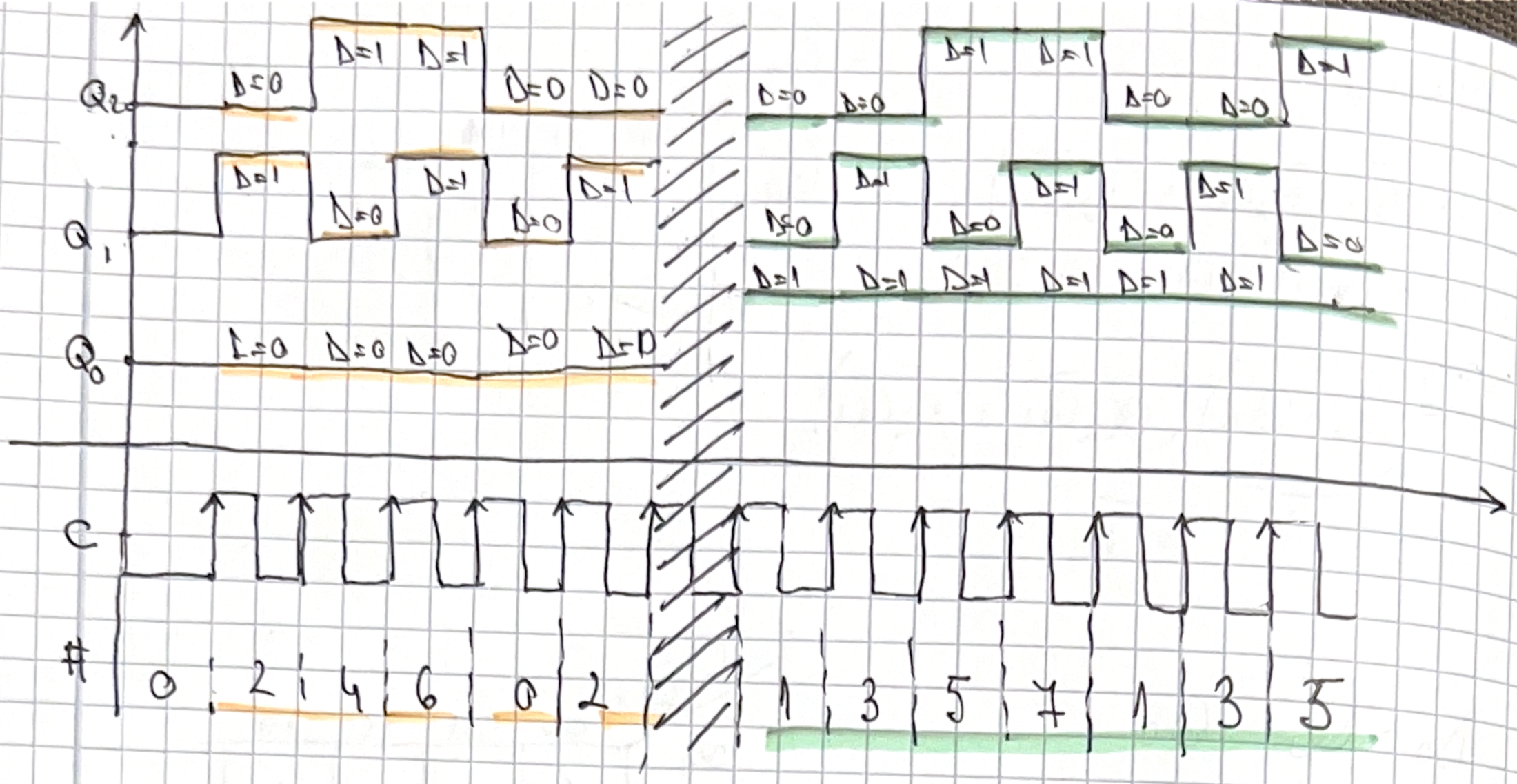
Q_0 \ Q_2Q_1	00	01	11	10
0	1	0	0	1
1	1	0	0	1

$$D_0 = Q_0' = \bar{Q}_0$$

Q_0 \ Q_2Q_1	00	01	11	10
0	0	0	0	0
1	1	1	1	1



Q_2	Q_1	Q_0	Q_2'	Q_1'	Q_0'	J_2	K_2	J_1	K_1	J_0	K_0
0	0	0	0	1	0	0	X	1	X	0	X
0	0	1	0	1	1	0	X	1	X	X	0
0	1	0	1	0	0	1	X	X	1	0	X
0	1	1	1	0	1	1	X	X	1	X	0
1	0	0	1	1	0	X	0	1	X	0	X
1	0	1	1	1	1	X	0	1	X	X	0
1	1	0	0	0	0	X	1	X	1	0	X
1	1	1	0	0	1	X	1	X	1	X	0



$$J_2 = (\overline{Q_2}Q_1 + Q_2\overline{Q_1})\,\overline{Q_0} + (\overline{Q_2}\,\overline{Q_1} + Q_2 Q_1)\,Q_0 =$$

Q_2Q_1 / Q_0

	00	01	11	10
1	0	1	×	×
0	0	1	×	×

$$= \underbrace{(Q_0 + \overline{Q_0})}_{1}(\overline{Q_2}Q_1 + Q_2\overline{Q_1})$$

$$= Q_1(\underbrace{\overline{Q_2} + Q_2}_{1}) = Q_1$$

$$K_2 = Q_1$$

Q_2Q_1 / Q_0

	00	01	11	10
0	×	×	1	0
1	×	×	1	0

$$J_1 = 1 = K_1$$

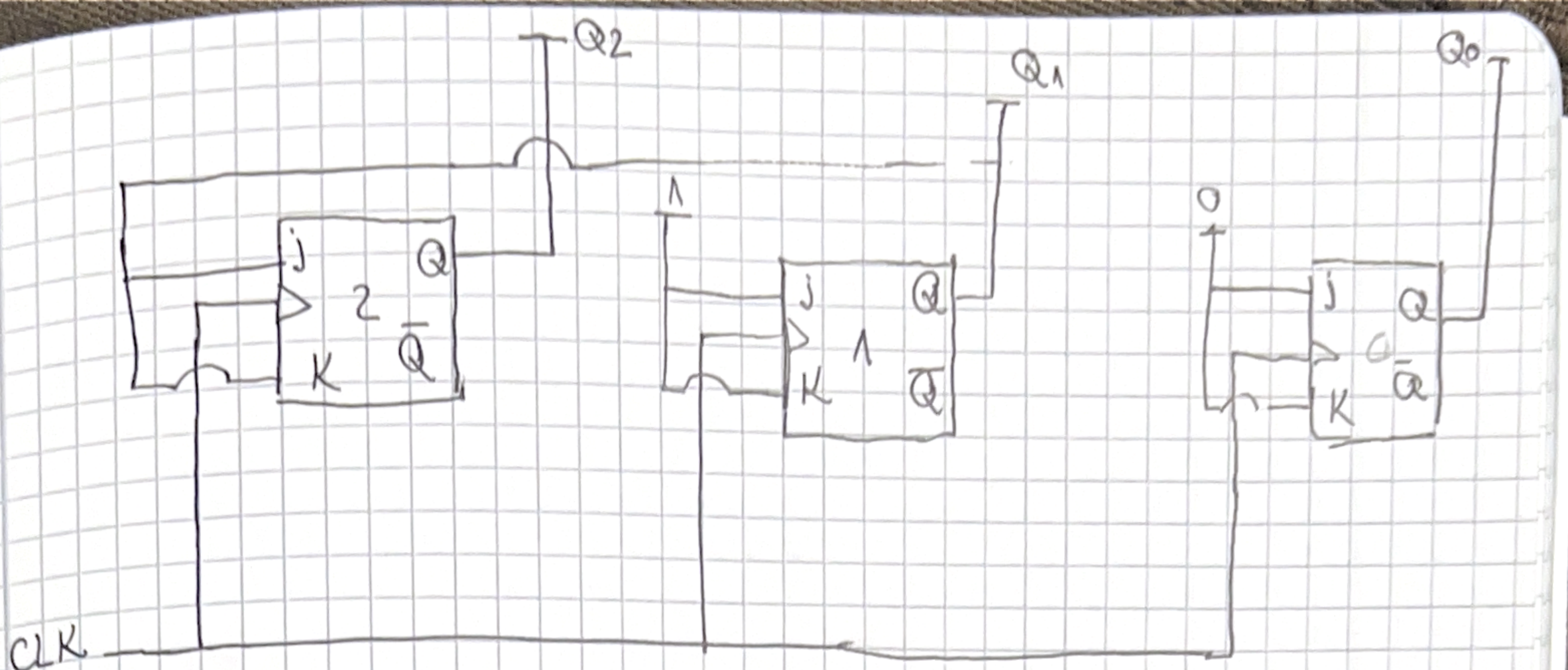
Q_2Q_1 / Q_0

	00	01	11	10
0	1	×	×	1
1	1	×	×	1

$$J_0 = K_0 = 0$$

Q_2Q_1 / Q_0

	00	01	11	10
0	0	0	0	0
1	×	×	×	×



$$D_2 = Q_2 \oplus Q_1$$

$$D_1 = \overline{Q_1}$$

$$D_0 = Q_0$$

$$J_2 = K_2 = Q_1$$

$$J_1 = K_1 = 1$$

$$J_0 = K_0 = 0$$

